

Phys 115B HW 8

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March 1, 2026

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Question 1. Consider a system with 2 N -state systems, with $N \gg 1$. For example, for the case where $N = 2$, you could imagine two spin $1/2$ particles, or for a case with $N = 100$, two particles in a 1-D harmonic oscillator but confined to the lowest 100 levels. The point is you have only two particles, but they can fill up to N levels. The system is assumed to be warm compared to ΔE , so all states are occupied equally.

- (a) What is the probability for finding the particles in the same state if they are distinguishable?
- (b) What is the probability for finding the particles in the same state if they are fermions?
- (c) What is the probability for finding the particles in the same state if they are bosons?

Proof. Let's fix the notation, say $\mathcal{H}$ is the N -dimensional Hilbert Space describing the N -state system, and $|\psi_1\rangle, \dots, |\psi_N\rangle$ are the allowed states upon observables.

- (a) If the two particles are distinguishable, then the Hilbert Space describing it is $\mathcal{H} \otimes \mathcal{H}$, so total of N^2 -dimensional. Which, there are N^2 distinct states (i.e. all $|\psi_i\rangle \otimes |\psi_j\rangle$, where $i, j \in \{1, \dots, N\}$), while N of them have both particles being in the same states (i.e. all $|\psi_i\rangle \otimes |\psi_i\rangle$, where $i \in \{1, \dots, N\}$). Hence, with each state being equally probable to observe, the probability the two distinguishable particles are in a same state is $\frac{N}{N^2} = \frac{1}{N}$.
- (b) If the two particles are fermions, they're not allowed to be in the same state, hence the probability of finding two fermions in the same state is 0.
- (c) If the two particles are bosons, then the Hilbert Space describing it is $S^2\mathcal{H}$ the symmetric tensor subspace in $\mathcal{H} \otimes \mathcal{H}$. Which, there are $\frac{N(N+1)}{2}$ distinct states (i.e. all $|\psi_i\rangle \otimes |\psi_j\rangle + |\psi_j\rangle \otimes |\psi_i\rangle$ for $i, j \in \{1, \dots, N\}$), while N of them have both particles being in the same states (i.e. all $|\psi_i\rangle \otimes |\psi_i\rangle$). Hence, with each state being equally probable to observe, the probability the two bosons are in a same state is $\frac{N}{N(N+1)/2} = \frac{2}{N+1}$.

□

Question 2. Consider a free electron gas with unequal numbers of spin-up and spin-down particles (N_+ and N_- respectively). Such a gas would have a net magnetization (magnetic dipole moment per unit volume)

$$M = -\frac{N_+ - N_-}{V} \mu_B \hat{k} = M \hat{k}$$

where $\mu_B = \frac{e\hbar}{2m_e}$ is the Bohr magneton (The minus sign is there because the charge of the electron is negative).

- (a) Assuming the electrons occupy the lowest energy levels consistent with the number of particles in each spin orientation, find E_{tot} . Check that your answer reduces to

$$E_{tot} = \frac{\hbar^2 (3\pi^2)^{5/3} (2N)^{5/3}}{10\pi^2 m} V^{-2/3}$$

when $N_+ = N_- = N$.

- (b) Show that for $M/\mu_B \ll \rho \equiv (N_+ + N_-)/V$ (which is to say $|N_+ - N_-| \ll (N_+ + N_-)$), the energy density is

$$\frac{1}{V} E_{tot} = \frac{\hbar^2 (3\pi^2 \rho)^{5/3}}{10\pi^2 m} \left[1 + \frac{5}{9} \left(\frac{M}{\rho \mu_B} \right)^2 \right].$$

The energy is a minimum for $M = 0$, so the ground state will have zero magnetization. However, if the gas is placed in a magnetic field (or in the presence of interactions between the particles) it may be energetically favorable for the gas to magnetize.

Proof.

- (a) Let $N_{tot} := N_+ + N_-$, and $N_{com} := \min\{N_+, N_-\}$ (so, how many electrons in total, and how many pairs of opposite spin electrons, respectively). Another way to represent the numbers is $N_{com} = \frac{N_+ + N_- - |N_+ - N_-|}{2}$.

Since each energy level allows two electrons with opposite spins, it's the most favorable for the lowest energy levels to be filled with electron pairs with opposite spins. Hence, for N_{com} pairs of electrons, they can be treated as cases with no net magnetization, hence agrees with the formula derived. Or, the energy they occupied is:

$$E_{com} = \frac{\hbar^2 (3\pi^2)^{5/3} (2N_{com})^{5/3}}{10\pi^2 m} V^{-2/3} \quad (1)$$

- (b)

□

Question 3. *The free electron gas model ignores the Coulomb repulsion between electrons. Because of the exchange force, Coulomb repulsion has a stronger effect on two electrons with anti-parallel spins (which behave in a way like distinguishable particles) than two electrons with parallel spins (whose position wave function must be antisymmetric - we saw this with the carbon atom). As a crude way to take into account Coulomb repulsion, pretend that every pair of electrons with opposite spin carry extra energy U , while electrons with the same spin do not interact at all; this adds $\Delta E = UN_+N_-$ to the total energy of the electron gas. As you will show, above a critical value of U , it becomes energetically favorable for the gas to spontaneously magnetize ($N_+ \neq N_-$); the material becomes ferromagnetic.*

- (a) Rewrite ΔE in terms of the density ρ and the magnetization M from the previous problem.
- (b) Assuming that $M/\mu_B \ll \rho$, for what minimum value of U is a non-zero magnetization energetically favored?

Proof.

(a)

(b)

□

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Question 4. Show that for a Hermitian operator $\hat{Q}$, the operator $\hat{U} = e^{i\hat{Q}}$ is unitary. Hint: first prove that the adjoint is given by $\hat{U}^\dagger = e^{-i\hat{Q}}$; then prove that $\hat{U}^\dagger \hat{U} = 1$.

Proof. Assume the exponentiation converges absolutely point wise (and that adjoint commutes with summation), then with $\hat{Q}$ being self adjoint (Hermitian), one have the following:

$$\hat{U} = e^{i\hat{Q}} = \sum_{n=0}^{\infty} \frac{1}{n!} (i\hat{Q})^n \quad (2)$$

In particular, the adjoint is given as follow:

$$\hat{U}^\dagger = \left(\sum_{n=0}^{\infty} \frac{1}{n!} (i\hat{Q})^n \right)^\dagger = \sum_{n=0}^{\infty} \frac{1}{n!} ((i\hat{Q})^\dagger)^n = \sum_{n=0}^{\infty} \frac{1}{n!} (-i\hat{Q})^n = e^{-i\hat{Q}} \quad (3)$$

Which, the two operators composes to the following:

$$\hat{U}^\dagger \hat{U} = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{1}{(n!)(m!)} (-i\hat{Q})^n (i\hat{Q})^m = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{1}{(k!)((n-k)!)} (-i\hat{Q})^k (i\hat{Q})^{n-k} \quad (4)$$

$$= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n \frac{(-1)^k}{(k!)((n-k)!)} \right) (i\hat{Q})^n \quad (5)$$

Which, for $n = 0$, the coefficient $\sum_{k=0}^n \frac{(-1)^k}{(k!)((n-k)!)} = 1$; else if $n > 0$, notice the following factor:

$$\forall z \in \mathbb{C}, \quad ((-1) + z)^n = \sum_{k=0}^n \binom{n}{k} (-1)^k z^{n-k} = \sum_{k=0}^n \frac{n!}{k! (n-k)!} (-1)^k z^{n-k} \quad (6)$$

Which, take $z = 1$, one arrives at $0 = ((-1)+1)^n = n! \sum_{k=0}^n \frac{(-1)^k}{k! (n-k)!}$, hence with $n! \neq 0$, then $\sum_{k=0}^n \frac{(-1)^k}{k! (n-k)!} = 0$.

As a result, we have:

$$\hat{U}^\dagger \hat{U} = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n \frac{(-1)^k}{(k!)((n-k)!)} \right) (i\hat{Q})^n = (i\hat{Q})^0 = 1 \quad (7)$$

Grinding out similar summation, one has $\hat{U} \hat{U}^\dagger = 1$, showing $\hat{U}^\dagger = \hat{U}^{-1}$, hence $\hat{U}$ is unitary. $\square$

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Question 5. When we first introduced Dirac notation, we fixed the inner product of position and momentum eigenstates using our foreknowledge of the position-space representation of states of definite momentum, i.e. we asserted that

$$\langle x|p\rangle = Ae^{ipx/\hbar}$$

where A is a normalization factor. Now we can derive this using the properties of the translation operator $T(a)$. If the state $|p\rangle$ is one of well-defined momentum p , how would you characterize state $|p'\rangle = T(a)|p\rangle$? Show that the position-space wavefunctions of these states are related by

$$\langle x|p'\rangle = \psi_{p'}(x) = e^{-iap/\hbar}\psi_p(x) = e^{-iap/\hbar}\langle x|p\rangle$$

and

$$\psi_{p'}(x) = \psi_p(x - a).$$

Then conclude that

$$\langle x|p\rangle = Ae^{ipx/\hbar}.$$

Proof. Assuming we know the relation $\langle x|\psi\rangle = \psi(x)$, and $\langle x|T(a)|\psi\rangle = \psi(x - a)$, while $T(a)^\dagger = T(a)^{-1} = T(-a)$. Also, one has $\hat{P}$ the momentum operator satisfies $T(a) = e^{-ia\hat{P}/\hbar} = \sum_{n=0}^{\infty} \frac{(-ia/\hbar)^n}{n!} \hat{P}^n$ (where $\hat{P}|p\rangle = p|p\rangle$ by definition of “definite momentum”, which is an eigenvector of the momentum operator). Hence, one has the following:

$$\langle x|p'\rangle = \langle x|(T(a)|p\rangle) = \langle x|\left(\sum_{n=0}^{\infty} \frac{(-ia/\hbar)^n}{n!} \hat{P}^n |p\rangle\right) \quad (8)$$

$$= \langle x|\left(\sum_{n=0}^{\infty} \frac{(-ia/\hbar)^n}{n!} p^n |p\rangle\right) = \langle x|\left(\sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{iap}{\hbar}\right)^n |p\rangle\right) \quad (9)$$

$$= \langle x|\left(e^{-iap/\hbar} |p\rangle\right) = e^{-iap/\hbar} \langle x|p\rangle \quad (10)$$

Hence, one derives that $\psi_p(x - a) = \psi_{p'}(x) = e^{-iap/\hbar}\psi_p(x)$ (given the definition $\langle x|T(a)|p\rangle = \psi_p(x - a)$). As a result, take $u := -a$ (where $a \in \mathbb{R}$ is arbitrary), and plugin $x = 0$, one has the following:

$$\psi_p(u) = \psi_p(0 - a) = e^{-iap/\hbar}\psi_p(0) = \psi_p(0)e^{ipu/\hbar} \quad (11)$$

Hence, with $A = \psi_p(0)$, so $\psi_p(u) = Ae^{ipu/\hbar}$ for all $u \in \mathbb{R}$. $\square$